

Report on Silicon Photonics Design, Fabrication and Data Analysis

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1. Abstract

Interferometers are important components in Photonic Integrated Circuits (PICs) since they are often used to implement modulators and switches. In addition, interferometers can be used to characterize the dispersion of optical waveguides. In this design project, passive interferometers are used to investigate waveguides and measure waveguide parameters such as the group index. Bragg gratings and racetrack resonators were also designed and measured.

2. Modeling and Simulation

2.1. Waveguides

The devices in this project are fabricated from silicon-on-insulator (SOI) wafers. The carrier wafer is 725 μm thick with a 2 μm layer of SiO_2 and a 220 nm of Si. The Si is etched to form the passive PIC components and another 2 μm layer of SiO_2 is deposited on the etched Si.

The standard waveguide width for 220nm Si in SiO_2 is 500 nm. If the waveguide width is too narrow, excessive scattering losses from sidewall roughness will result since the smaller width spreads more of the mode into the cladding. However if the width is too wide, an excess number of modes will be present. Imperfections in components, in addition to sidewall roughness, can lead to mode conversion. This will deteriorate PIC device performance, since the effective index and group velocity of each mode are different.

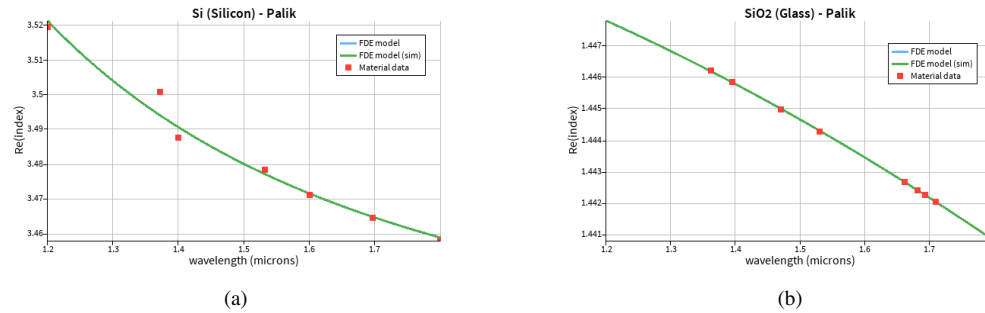


Fig. 1. Fitted material dispersion for a) Si and b) SiO_2

Eigenmode simulations of the Si waveguides were performed in Lumerical Mode. The first step is to include the material dispersion of Si and SiO_2 in the simulation. As seen in Figure 1, the index of Si from 1500 nm to 1600 nm varies by approximately 0.01. Although this index change is seemingly small, the length of the interferometers will be on the order of hundreds or thousands of wavelengths. Thus it is important to take material dispersion into account. The variation of SiO_2 is approximately an order of magnitude less. This is expected from the Kramers-Kronig relations, as silicon exhibits strong absorption near 1100nm, whereas SiO_2 is transparent at optical wavelengths.

Plots of the E-field amplitude $|E(x, y)|$ are shown for the first three modes at 1550 nm in Figure 2. The first mode (the TE_1 mode) shown in Figure 2 a) and d) is quasi-TE polarized (TE fraction 98%) with effective index $n_{\text{eff}} = 2.44$. The discontinuity in the E-Field at the Si- SiO_2 interface is

due to the index discontinuity and the normal boundary condition. [At the boundary the normal component of the Displacement field $\mathbf{D} = \epsilon(x, y)\mathbf{E}$ and the tangential component of the Electric field are continuous.] Note that the E-field discontinuity is greatest at the vertical sidewalls since the electric field of TE_1 mode is primarily polarized in the horizontal or x -direction.

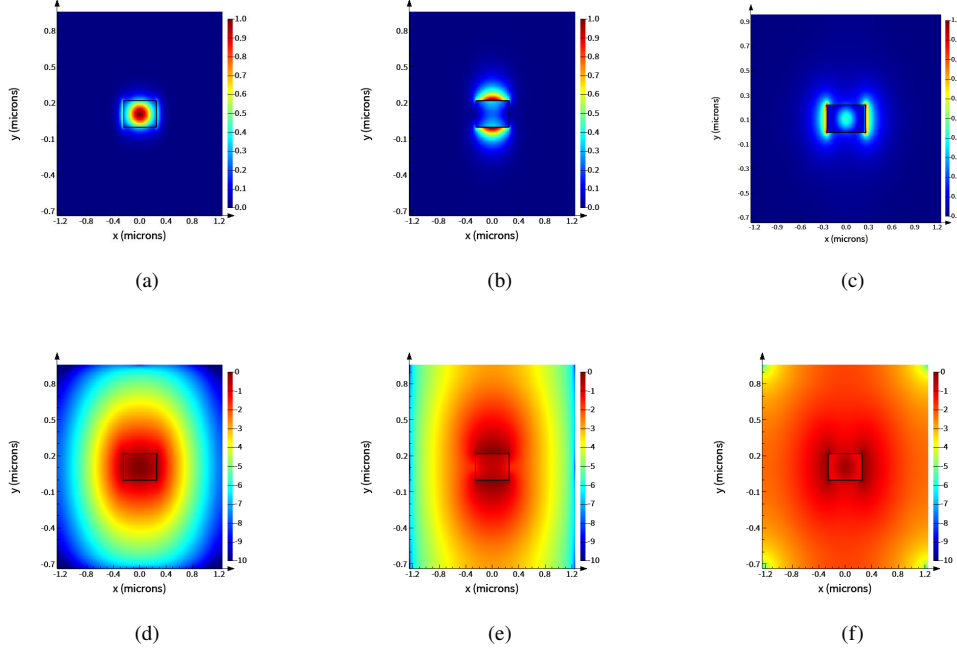


Fig. 2. Amplitude of electric field at 1550 nm for a) first TM mode b) first TE mode c) second TM mode. d-f) corresponding plots on log scale.

In contrast to the TE_1 mode, the plots of the TM_1 mode E-field show a discontinuity on the horizontal sidewalls. This is due to the TM_1 mode being primarily polarized in the vertical or y -direction. The effective index of the TM_1 mode ($n_{\text{eff}} = 1.77$) is less than the TE mode. As a consequence the TM_1 mode is more extended in the SiO_2 cladding region as shown in Figure 2 e). Finally there is a 3rd hybrid mode which is 65% TE polarized shown in Figure 2 c) and f). This mode has strong E-field discontinuity at the vertical sidewalls and appreciable E-field discontinuity at the horizontal sidewalls near the corners. Its effective refractive index ($n_{\text{eff}} = 1.49$) is close to the cladding SiO_2 . Consequently the mode is lightly confined to the core region with the majority of the mode in the cladding region. Any sidewall roughness should strongly scatter this mode into the radiation modes. The scattering losses for this hybrid mode should be much greater than the TE_1 .

The effective index and group index of the TE_1 mode are shown in Figure 3. A wavelength sweep of the TE_1 mode from 1500 nm to 1600 nm was performed with 20 points. Note that variation of the effective index is ≈ 0.1 . This is one order of magnitude greater than the material dispersion of Silicon. Thus the waveguide dispersion is much greater than the material dispersion. The waveguide effective index decreases with increasing wavelengths, since a larger wavelength forces more of the mode to be outside the core.

The variation of the group index in Figure 3 b) is much less than the variation in the effective index. Here the dispersion of the second term $-\lambda \frac{dn}{d\lambda}$ is found to roughly compensate for the first term $n_{\text{eff}}(\lambda)$ in eq. 11. The n_{eff} data are fitted to a 2nd order polynomial. The least squares fit is

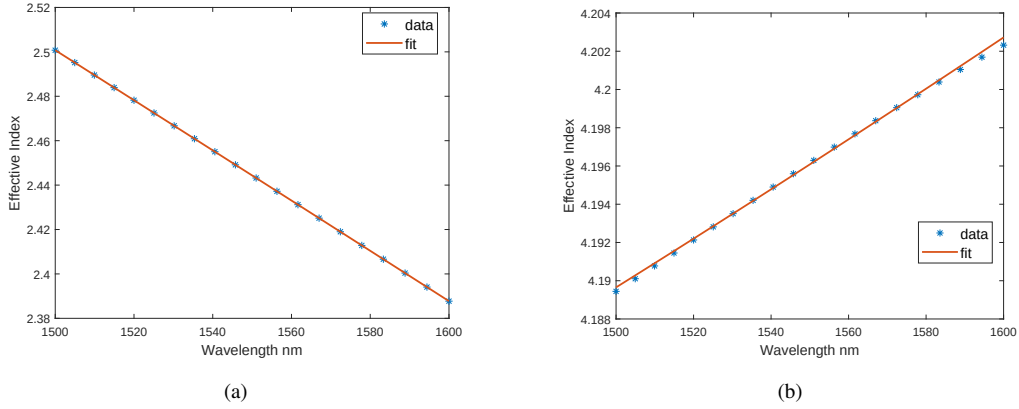


Fig. 3. a) Effective index and b) group index of the waveguide

$$n_{\text{eff}}(\lambda) = 2.44435 - 1.13015(\lambda - \lambda_o) - 0.0421608(\lambda - \lambda_o)^2 \quad (1)$$

where the units of λ_o - 1.55 μm . By taking the derivative of $n_{\text{eff}}(\lambda)$ and using eq. 11 we can use the polynomial fit of n_{eff} to obtain an polynomial expression for n_g . As shown in Figure 3 b) the polynomial expression for n_g fits the data well but there are discrepancies especially for wavelengths near 1500 nm and 1600 nm. My assumption is that Lumerical Mode obtains the group velocity and index of each modes by other methods and not by fitting the effective index data.

2.2. Interferometer

Consider a simple interferometer, where light is first split between two paths of different lengths and refractive indices and then recombined. We assume that light travels each path once as in a Mach-Zehnder interferometer and not twice as in the Michelson Morley interferometer. Now the frequency along each path remains the same, but index and hence wavelength of each path can be different. We make the problem one-dimensional and write the electric field in the two arms of the interferometer as

$$\tilde{E}_1 = \text{Re} \{E_1 \exp\{+jk_1 L_1 - j\omega t\}\} \quad \tilde{E}_2 = \text{Re} \{E_2 \exp\{+jk_2 L_2 - j\omega t\}\} \quad (2)$$

where L_1 and L_2 are the lengths of each path, ω is the angular frequency, t is the time, $j = \sqrt{-1}$, E_1 and E_2 are the complex amplitudes and the tilde represents the physical E-field. At the output the detectors measure the optical power which is proportional to the square of the physical field. To find $(\tilde{E}_1 + \tilde{E}_2)^2$ denote the complex fields as $z_i \equiv \text{Re}(E_i \exp\{+jk_i L_i - j\omega t\})$ and note that for any complex number $\text{Re}\{z\} = (z + z^*)/2$

$$\begin{aligned} (\tilde{E}_1 + \tilde{E}_2)^2 &= \left[\left(\frac{z_1 + z_1^*}{2} \right) + \left(\frac{z_2 + z_2^*}{2} \right) \right]^2 = \frac{1}{4} \left[(z_1 + z_1^*)^2 + (z_2 + z_2^*)^2 + 2(z_1 + z_1^*)(z_2 + z_2^*) \right] \\ &= \frac{1}{4} \left[(z_1^2 + z_1^{*2}) + 2|z_1|^2 + (z_2^2 + z_2^{*2}) + 2|z_2|^2 + 2(z_1 z_2 + z_1^* z_2^*) + 2(z_1^* z_2 + z_1 z_2^*) \right] \end{aligned} \quad (3)$$

Numbers and their corresponding complex conjugates are grouped together in parenthesis. All the terms in parenthesis are thus real numbers and correspond to physical terms. The following terms $(z_1^2 + z_1^{*2})$, $(z_2^2 + z_2^{*2})$, $(z_1 z_2 + z_1^* z_2^*)$ have $e^{\pm j2\omega t}$ factors. Their time dependence will

be proportional to $\cos(2\omega t)$. These oscillation at nearly 400 THz are too fast for the detector and will average to zero. The other terms $|z_1|^2$, $|z_2|^2$, $(z_1^* z_2 + z_1 z_2^*)$ have no time dependence since the product of exponential terms always equals unity, $e^{-j\omega t} e^{+j\omega t} = 1$. Thus the detector response will be

$$\frac{1}{2}|E_1|^2 + \frac{1}{2}|E_2|^2 + |E_1||E_2| \cos [k_1 L_1 - k_2 L_2 + \phi_1 - \phi_2] \quad (4)$$

where ϕ_i is the phase of the complex amplitude E_i . If we assume the complex amplitudes and the refractive index of each waveguide are equal such that $E_1 = E_2 \equiv E_o$, then

$E_{in}^2 = (E_1^2 + E_2^2)$ the transfer function of the interferometer can be written as

$$\boxed{\frac{I_{out}}{I_{in}} = \frac{E_{out}^2}{E_{in}^2} = \left(\frac{1 + \cos [k\Delta L]}{2} \right)} \quad (5)$$

where $\Delta L = L_1 - L_2$. Maximum output occurs when $k\Delta L = 0, 2\pi, 4\pi \dots$ while minimum output occurs when $k\Delta L = \pi, 3\pi \dots$. The wave-vector k can be viewed as either a function of wavelength λ , angular frequency ω or frequency ν .

$$k(\omega) = \frac{n(\omega)\omega}{c} \quad k(\lambda) = \frac{2\pi n(\lambda)}{\lambda} \quad k(\nu) = \frac{2\pi n(\nu)\nu}{c} \quad (6)$$

For derivations it is easier to use $k(\omega)$. However measurements are taken as a function of wavelength. Therefore we make all derivations in terms of $k(\omega)$ and then convert them to λ . For conversions the following relations are useful

$$c = \lambda\nu \quad \omega = ck \quad \omega = \frac{2\pi c n}{\lambda} \quad (7)$$

Note that λ is the always the wavelength in vacuum. In a medium with refractive index n length of the wave will be reduced to λ/n which will decrease the effective phase velocity of light to c/n .

2.3. Group Velocity

In addition to the phase velocity in a dispersive material (where n varies with λ) the energy of the electromagnetic wave travels at the group velocity. If the envelope of a light pulse varies slowly with respect to the field oscillations, the group velocity corresponds to the speed of the pulse-envelope, while the phase velocity corresponds to the speed of the field oscillations. In the absence of dispersion the group and phase velocities are equal.

The phase v_p and group velocity v_g and corresponding phase index n and group index n_g are defined as

$$v_p = \frac{\omega}{k} = \frac{c}{n(\omega)} \quad v_g = \frac{d\omega}{dk} = \frac{c}{n_g(\omega)} \quad (8)$$

The phase and group index are then

$$n(\omega) = c \left(\frac{k}{\omega} \right) \quad n_g(\omega) = c \left(\frac{dk}{d\omega} \right) \quad (9)$$

Using the expression for $k(\omega)$ of eq. 6 and substituting it into the expression for $n_g(\omega)$ in eq. ?? and using the chain-rule gives an expression for $n_g(\omega)$ in terms the dispersion in ω .

$$n_g(\omega) = c \frac{dk(\omega)}{d\omega} = c \frac{d}{d\omega} \left(\frac{n(\omega)\omega}{c} \right) = n(\omega) + \omega \frac{dn(\omega)}{d\omega} \quad (10)$$

For the measurements and data the units to be in wavelength λ . So we just need use the relation between ω and λ in eq. 7.

$$\lambda = \frac{2\pi cn}{\omega} \Rightarrow \frac{d\lambda}{d\omega} = -\frac{2\pi cn}{\omega^2} \quad \frac{dn}{d\omega} = \left(\frac{dn}{d\lambda}\right)\left(\frac{d\lambda}{d\omega}\right) \Rightarrow \boxed{n_g(\lambda) = n(\lambda) - \lambda \frac{dn}{d\lambda}} \quad (11)$$

Since the effective index of the Si waveguides decreases with wavelength the sign of $\frac{dn}{d\lambda}$ is negative. Therefore the group index will be larger than the phase index since the negative sign in the expression of eq. 11 will make the last term positive.

2.4. Free Spectral Range

The separation between the maximums of the interferometer transfer function of eq. 5 is known as the free spectral range (FSR). For consecutive maxima $k(\omega_{m+1}) - k(\omega_m) = 2\pi/\Delta L$. If the separation between the maxima is small then $k(\omega_{m+1}) - k(\omega_m) = \Delta k$ and $\omega_{m+1} - \omega_m = \Delta\omega$

$$\Delta k(\omega) \approx \frac{dk(\omega)}{d\omega} \Delta\omega = \frac{2\pi}{\Delta L} \Rightarrow \Delta\omega_{FSR} = \frac{2\pi v_g}{\Delta L} = \frac{2\pi c}{n_g \Delta L} \Rightarrow \Delta\nu_{FSR} = \frac{c}{n_g \Delta L} \quad (12)$$

We can then convert this expression to wavelength

$$\nu = \frac{c}{\lambda} \Rightarrow \frac{d\nu}{d\lambda} = -\frac{c}{\lambda^2} \Rightarrow \Delta\nu \approx \frac{c}{\lambda^2} \Delta\lambda \Rightarrow \boxed{\Delta\lambda_{FSR} = \frac{\lambda^2}{n_g \Delta L}} \quad (13)$$

With regards to the minus signs note that $\Delta\omega$ and Δk have different signs and a more careful derivation would take this into account.

2.5. Effective waveguide index

For light propagating in a uniform three-dimensional medium the direction of travel is defined by the wave-vector $\mathbf{k}$ whose magnitude $k \equiv |\mathbf{k}|$ is used in the above equations. However for optical waveguides light is confined in two transverse directions (x and y) by total internal refraction. The index of central region or core of the waveguide must be greater than the surrounding index of the cladding $n_{co} > n_{cl}$. In the transverse directions interference forms standing wave patterns. In the longitudinal direction (z) the wave is free to propagate. The longitudinal component of the wave-vector is denoted by $\beta \equiv k_z$. The effective index of the waveguide n_e is defined from $\omega = (c/n_e)\beta$. Since $|k| = \sqrt{k_x^2 + k_y^2 + \beta^2}$ the effective index of the waveguide is always less than that of the core. We thus expect higher order modes to have lower effective indices since larger transverse spatial frequencies are required to produce more nodes. Note that the above sections on interferometers can be applied to waveguides by simply replacing the 1-D wave-vector k with the propagation constant β and the index n with the effective index n_{eff} which include both material and waveguide dispersion.

Width / Height	470 nm	490 nm	510 nm
215.3 nm	4.246	4.209	4.175
219.2 nm	4.251	4.214	4.179
223.1 nm	4.255	4.217	4.182

Table 1. Corner analysis of group index for different waveguide widths and heights.

2.6. Corner Analysis

Although, the nominal values of the waveguide height and width are 220nm and 500nm, values of the fabricated waveguide will vary. To investigate the impact of these variations a corner analysis is performed at 1.55 μm . Since the group index can be directly obtained from the measured FSR of the interferometers, only the group index corner analysis is presented. In contrast to the group index, the FSR does not depend on the effective index.

From previous fabrication runs, the minimum and maximum waveguide widths were 470 and 510 nm, respectively, for a nominal width of 500 nm. The SOI supplier reported a mean thickness of 219.2nm with a standard deviation of 3.9 nm. From the corner analysis a general trend emerges that for greater widths the group index decreases while for greater heights the group index increases.

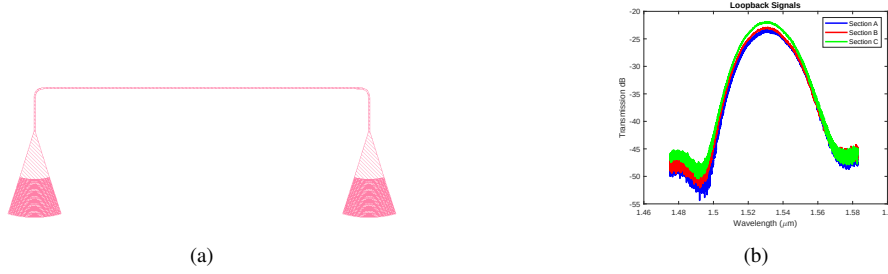


Fig. 4. a) Loopback structure consisting of two grating couplers and connecting waveguide. b) Transmission of loopback structures.

3. Interferometers

3.1. Loopbacks structures

Light from the fiber-optic array was coupled into the devices using grating couplers [1]. The grating couplers are both sensitive to wavelength and fiber angle. A schematic of a "loopback" structure used to characterize grating couplers is shown in Figure 4 a). The transmission for the loopback structure for each design section is shown in Figure 4 b). The nominal design wavelength is 1.55 μm but the maximum transmission is at 1.53 μm . This is most likely due to fabrication variances or a slight angular misalignment of the fiber angle with respect to the Si surface. Since there are two grating couplers in each loopback structure, the coupling loss of each grating coupler is on the order of 10dB. Note the waveguide loss which is on the order of 5 dB/cm should be negligible for these measurements.

3.2. Interferometer series

A series of interferometers was produced with path length differences of 37.5 μm , 75 μm , 150 μm , 300 μm , 600 μm and 1200 μm . The path length differences were chosen so that the free spectral range of devices differed by factors of two. The interferometers employed two Y-Branch splitters [2] to separate and recombine the beams. Two sets of interferometers were produced. One set has a total of 5 turns of 90° bends in each interferometer arm [as shown in Figure 5a) and b)] and the other set as a total of 22 turns in each interferometer arm. For the 5 turn interferometer series the nominal 600 μm device was mistakenly designed with a path length difference of 560 μm .

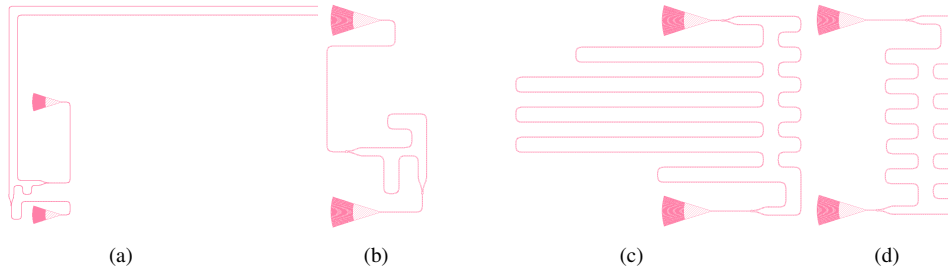


Fig. 5. Interferometers with 5 turns on each arm with 1200 μm a) and 37.5 μm b) path length difference. Interferometers with 22 turns on each arm with 1200 μm c) and 37.5 μm d) path length difference.

3.3. Data Analysis from finding fringe minima and fitting

The transmission from an interferometer with a path length difference of 1200 μm is shown in Figure 6a). The maximum in the data at 1.53 μm is due to the response of the grating couplers. To remove the grating coupler response the raw transmission data is fit to a 4th order polynomial to find a baseline. The baseline is then used to correct the data - Figure 6b). Alternatively the loopback data in 5 can be used for data correction. Gaussian convolution is then used to smooth the corrected data Figure 6c). After smoothing the wavelengths at which the fringe minima occur are found - Figure 6d). The free spectral range (FSR) can then be calculated - 6d), The variation of the FSR with wavelength is indicative of dispersion. From eq. 13 the group index n_g can be determined from the FSR. The effective index n_{eff} is then modeled as a 2nd order polynomial

$$n_{\text{eff}} = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \quad (14)$$

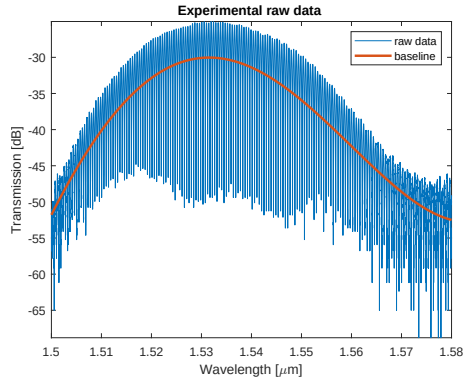
The value of n_1 is estimated to be 2.4 and then adjusted so that one of the fringe minima corresponds to an integer mode number for λ_0 . Here λ_0 is chosen to be roughly in the center of the wavelength range. The group velocity is then approximated as

$$n_g \approx n_1 - \lambda_0 n_2 \quad (15)$$

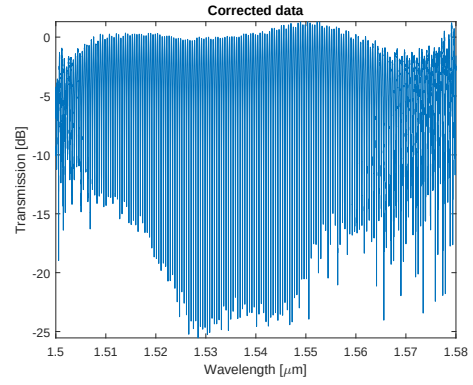
A rough estimate of n_2 can then be obtained from the mean measured value of n_g in Figure 6f), the known value of λ_0 and the assumed value of n_1 . In theory the value of n_3 form the derivative of the group index with respect to wavelength. However this value is found to fluctuate greatly, hence we estimate n_3 to be zero. The estimates of n_1 , n_2 and n_3 are then used as initial values for a non-linear least square fit of the smooth corrected data - Figure 6f). With the fitted coefficients n_1 , n_2 and n_3 known the group index can be calculated by numerically differentiating $n_{\text{eff}}(\lambda)$ and using eq. 11 to find the group velocity. The result is n_{eff} as shown in Figure 6g).

If the parameter n_1 is significantly changed from the nominal value of 2.4 the non-linear fitting of effective index is still very good. For example if we choose the initial estimate $n_1 = 2.4$. The group index at 1.55 μm is 4.1876. While if $n_1 = 3.4$ is chosen the group index at 1.55 μm is 4.868 - almost identical! Even though the polynomial best fit coefficients for the $n_1 = 2.4$ initial estimate [$n_1 = 2.3867$, $n_2 = -1.1618 \mu\text{m}^{-1}$, $n_3 = 0.0024433 \mu\text{m}^{-2}$] are significantly different from best fit coefficients for the $n_1 = 3.4$ initial estimate [$n_1 = 3.3955$, $n_2 = -0.51052 \mu\text{m}^{-1}$, $n_3 = 0.0024433 \mu\text{m}^{-2}$]. Thus the values of the fits are not very meaningful. In addition we insert the polynomial expression for the effective index eq. 15 into the expression for the group index eq. 11. Then we obtain the following expression without linear terms in λ for group index.

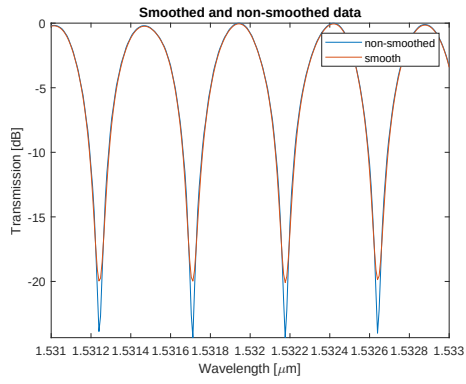
$$n_g = (n_1 + \lambda_0 n_2) + n_3 \lambda^2 \dots \quad (16)$$



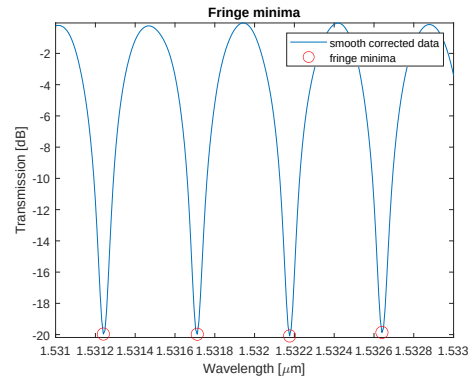
(a)



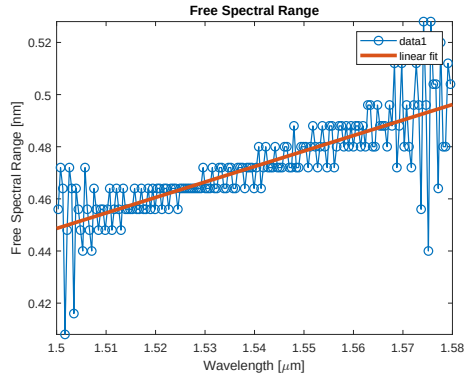
(b)



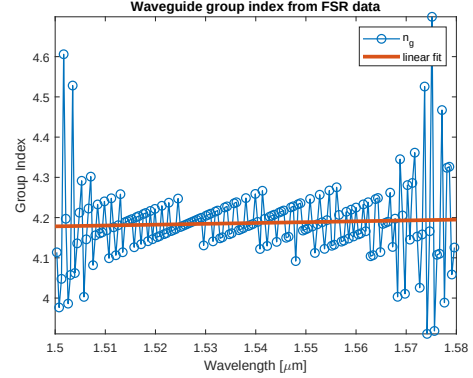
(c)



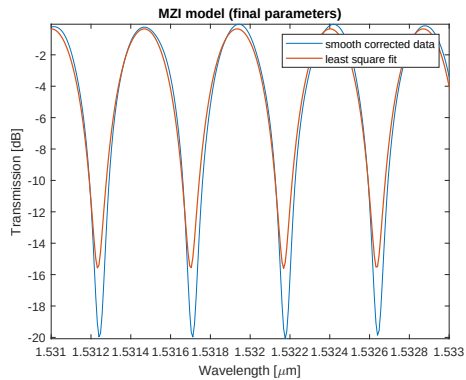
(d)



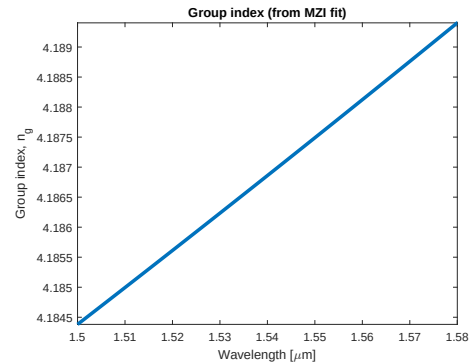
(e)



(f)



(g)



(h)

Fig. 6. a-h) Various steps in the analysis of the interferometer data.

If higher order polynomials are used to fit the effective index then there will linear terms in n_g that depend on n_4, n_5 etc. This is rather paradoxical as the group index n_g extracted from the FSR data has slope that is consistently positive. However the variation in n_g with λ is rather small. In addition on closer inspection of the 2nd order polynomial fit to the numerically computed n_g versus wavelength in Figure 3b) we see a clear pattern in the residuals. The polynomial fit is consistently high on the left and right side of the curve while being consistently low for the middle portions. Thus a second order polynomial may not be ideal for fitting the effective index.

3.4. Interferometer results

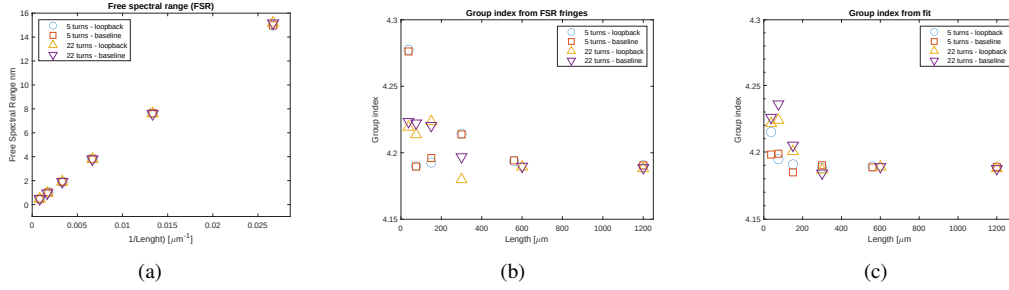


Fig. 7. a) Free spectral range (FSR) versus inverse path length difference. Group index found from b) FSR and non-linear fit c).

The free spectral range (FSR) and group index for all interferometers are shown in Figure 7. Note that for part a) the x-axis is the reciprocal of the path length difference. As expected the FSR is found to roughly double when the path length difference is halved. The group index - Figure 7 b) and c) - shows more variations for smaller length differences. Large path length differences have many fringes which will average out variations. In contrast smaller path length differences have fewer fringes and are hence more susceptible to variation.

For all the data the group index at $1.55 \mu\text{m}$ found from the FSR is $n_g = 4.207 \pm 0.025$ while the group index at $1.55 \mu\text{m}$ found from the non-linear polynomial fits is $n_g = 4.198 \pm 0.015$. Since the interferometers with the longest path length are expected to be the most accurate we give the mean group index from the FSR $n_g = 4.194$ and the non-linear polynomial fitting $n_g = 4.188$.



Fig. 8. a) Dispersion derived from free spectra range data versus path length difference b) Dispersion derived from non-linear polynomial fits

The dispersion D is defined as the derivative of the inverse group velocity with respect to wavelength.

$$D \equiv \frac{d}{d\lambda} \left(\frac{n_g(\lambda)}{c} \right) \quad (17)$$

The dispersion determines the spreading of an electromagnetic pulse (composed of different wavelength) in time. If the effective index is described by the polynomial in eq. 15 then the Dispersion is $2n_3/c$. The dispersion can also be calculated directly from the FSR by finding the group index and then numerically taking the derivative.

The dispersion is plotted versus the path length difference in Figure 8. The dispersion is plotted on a logarithmic scale due to the large variance in values. (There were even a negative values and the absolute value of the data is taken in Figure 8) The dispersion obtained from the FSR is $D = 1631 \pm 3615$ (ps/nm/km) while the dispersion obtained from the non-linear fits is $D = 2228 \pm 3186$ (ps/nm/km). For both cases the standard deviation is greater than the mean values. If we only consider the longest path length interferometers (1200 μm) the dispersion is found to be 222 (ps/nm/km) for the fits and 498 from the FSR data.

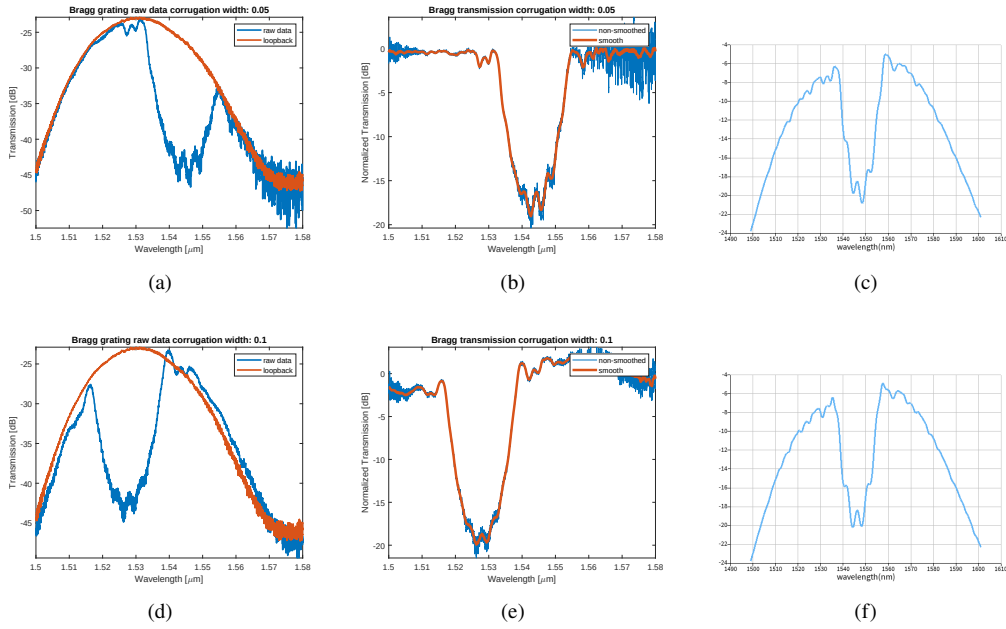


Fig. 9. Raw Bragg grating transmission and Loopback structure for corrugation width 0.05 μm a) and 0.1 μm d). Normalized Bragg grating transmission for corrugation width 0.05 μm b) and 0.1 μm e) Lumerical interconnect simulation of Bragg grating for corrugation width 0.05 μm c) and 0.1 μm f)

4. Bragg Gratings

Bragg grating waveguides have periodic variations in the width of the waveguide. The periodic variation in width from the nominal width is known as the corrugation length. Bragg gratings were fabricated with corrugation lengths of 0.05 μm and 0.1 μm in a simple loopback structure to measure the transmission. In hindsight, a Y-branch splitter should have been used to measure the reflection. The period of the grating was 0.317 μm and the total number of periods was 100. The experimental data along a loopback structure used for normalization is shown in Figure 9a) and d). The -3dB/(-10dB) width of the stopband for the 0.05 μm corrugation was 20nm/(15nm) with the central wavelength at 1.544 μm . For the 0.1 μm corrugation the -3dB/(-10dB) width of the stopband was 19nm/(16nm) with a central wavelength of 1.528 μm . This shift in the central wavelength is significantly different from the Lumerical interconnect shown in Figure 9 c) and f). The reasons for this shift are not clear, since no SEM pictures were taken which would have

revealed any manufacturing variations of the grating from the designed structures.



Fig. 10. Racetrack resonator with grating coupler. Leftmost grating coupler: drop/Channel 1. Middle left grating coupler: input. Middle right grating coupler: transmission/Channel 2. Rightmost grating coupler: reflection/Channel 4.

5. Resonators

A series of racetrack resonators were fabricated with varying gap sizes. The layout plane of the resonator is shown in Figure 10. The coupler length for the resonators is $10\ \mu\text{m}$ and the radius of the 90° curves was $5\ \mu\text{m}$. The length between the two coupler elements was chosen so that the round-trip path was $100\ \mu\text{m}$. The transmission signal and the "drop" signal in addition to what we term the "reflection signal" were measured. Ideally at a resonance of the racetrack filter there should be a sharp drop in the transmission. The optical power at the resonant frequency (which is not transmitted) should then couple to the opposite waveguide in the "drop" direction. However, there will be some reflections in the racetrack resonators at the boundary of $5\ \mu\text{m}$ radius curve waveguide section and the straight waveguide sections. These reflections lead to optical power being coupled to the opposite of the "drop" direction.

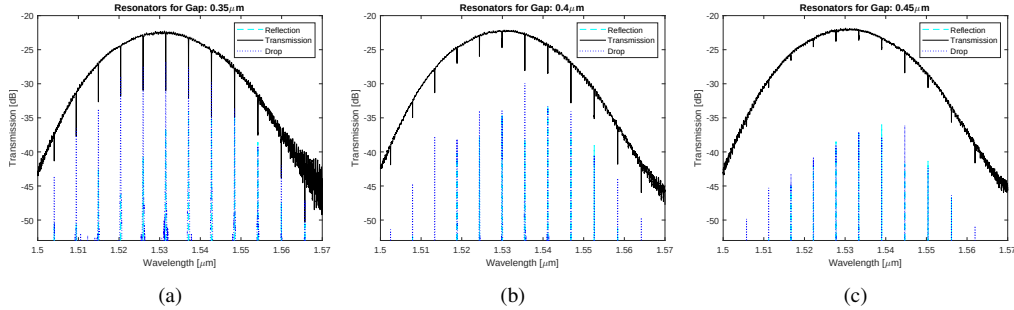


Fig. 11. Transmission for resonators with a) $0.35\ \mu\text{m}$ b) $0.40\ \mu\text{m}$ and c) $0.45\ \mu\text{m}$ gaps.

The resonator data for the $0.35\ \mu\text{m}$, $0.40\ \mu\text{m}$ and $0.45\ \mu\text{m}$ gaps are shown in Figure 11. The data for the $0.50\ \mu\text{m}$ gap resonator is not shown as it is similar to the $0.45\ \mu\text{m}$ gap resonator. For $0.35\ \mu\text{m}$ gap resonators the peak of the drop signal is nearly equal to the valleys of the transmission signal. Optical power is thus roughly conserved. Although there is a significant reflection signal, the y-axis is on a dB scale, and the reflection is signal typically smaller than the drop signal. The peaks and dips for $0.45\ \mu\text{m}$ and $0.5\ \mu\text{m}$ gaps do not meet. However, power at the resonance frequencies could be coupled into radiation modes in the $5\ \mu\text{m}$ radius curve waveguide section.

From the resonance peaks and dips the free spectral range of the resonators can be extracted from the data in a similar manner to the interferometers. Figure 12a) shows raw transmission data from the $0.4\ \mu\text{m}$ gap resonator structure. The raw data is smoothed using a Savitzky-Golay filter which preserves the peak heights. The peaks are then found from the smoothed data - Figure 12b). From the wavelengths and spacing of the peaks the free spectral range can be found as

shown in Figure 12c). In general we find the variance of the FSR versus wavelength data of the resonators to be much less than that of the interferometers. This might be due to the sharpness and delta-like features of the peaks as opposed to the oscillations of the interferometers.

From the FSR versus wavelength data the group index n_g can be extracted using eq. 13. The $n_g(\lambda)$ curve can then be numerically differentiated to find the dispersion. Figure 13 shows the group index and dispersion plotted versus the Resonator gap. The group index and dispersion are found to be $n_g = 4.1915 \pm 0.007$ and $D = 379 \pm 281$ (ps/nm/km).

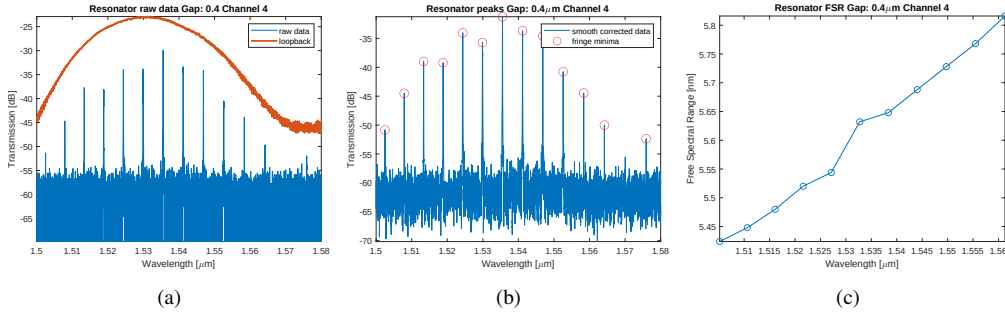


Fig. 12. a) Raw data and Loopback data b) Smoothed data with selected peaks c) Free spectral range

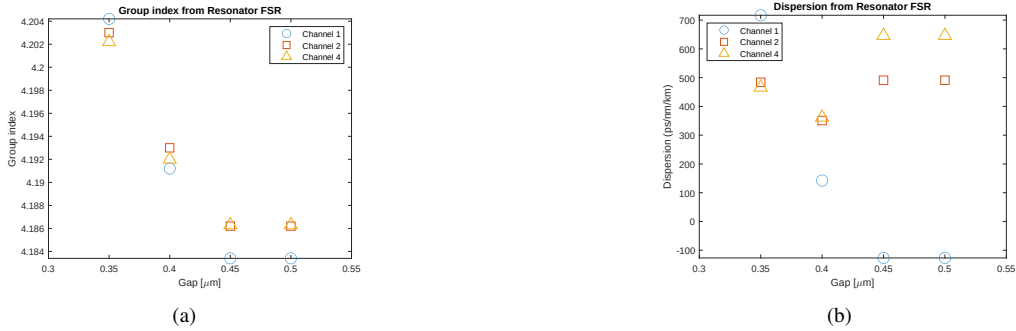


Fig. 13. a) Group index and b) dispersion of resonators

6. Conclusion

The table below shows a summary of the group index n_g and dispersion D obtained by various methods. For the resonators non-linear polynomial fits were not made hence the absence of data. For the 1200 μm interferometers there was not enough data points for standard deviation to have any meaning hence the absence of errors. All of the measured values for n_g were between the maximum and minimum of the corner analysis in Table 1. The error for the resonators was significantly less than that of the interferometers. This is probably due to the large variance of interferometers parameters for small path length difference. The interferometer data show that variances in n_g decrease with increasing path difference. Thus longer interferometers are more advantageous for measuring waveguide parameters. Since the longest path length had approximately 60 points between fringes it should be possible to increase the path length difference by at least a factor of 4 for improved n_g measurements. It should be noted that the loopback loss of the two grating couplers was abnormally high ($>20\text{dB}$). The low signal level could thus have contributed to the poorer performance of the short path length interferometers.

The uncertainties in the dispersion were much larger than the group index. With regard to the dispersion the best that can be concluded is that it is most likely on the order of several picoseconds.

Device	n_g (FSR)	n_g (fit)	D (FSR)	D (fit)
All Interferometers	4.207 ± 0.025	4.198 ± 0.015	1631 ± 3615	2228 ± 3186
1200 μm Interferometers	4.194	4.188	498	222
All Resonators	4.1915 ± 0.007		379 ± 281	

Group index and dispersion at 1.55 μm . The units for dispersion are (ps/nm/km).

7. Measurement Details

To characterize the devices, a custom-built automated test setup [3] [4] with automated control software written in Python was used [5]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [2]. A 90° rotation was used to inject light into the TM grating couplers [2]. A polarization maintaining fibre array was used to couple light in/out of the chip [6].

8. Fabrication Details

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc.(<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 μm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H₂SO₄:H₂O₂) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 μm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300 C°. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

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